"*AUTONOMOUS COMPUTATIONAL INTELLIGENCE FOR OVERSEEING THE HAZARDOUS INCIDENTS IN RAIL TERMINALS USING MACHINE LEARNING*"

A Dissertation submitted in partial fulfillment of the requirements for the award of a degree of

MASTER OF COMPUTER APPLICATIONS

Of

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Submitted By

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(1AM23MC056)

Under the guidance of

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18th KM, Bannerghatta Road, Bengaluru - 560083.

September 2025

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This is to certify that **SOUJANYA B (1AM23MC056)** *has completed her final Semester project work entitled* **"AUTONOMOUS COMPUTATIONAL INTELLIGENCE FOR OVERSEEING HAZARDOUS INCIDENTS IN RAIL TERMINALS USING MACHINE LEARNING"** *as partial fulfillment for the award of* ***Master of Computer Applications*** *degree, during the academic year of 2024-25 under our joint-supervision*

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The project was carried out during the academic year 2024-2025, and the performance of the student has been found to be satisfactory.

We wish her all the best and success in all future endeavors.

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Student, Department of MCA, AMC Engineering College, Bengaluru, India

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DECLARATION

I, **SOUJANYA B,** student of 4th semester MCA, **AMC Engineering College**, bearing USN **1AM23MC056** here by declare that the project entitled "**Autonomous Computational Intelligence for Overseeing Hazardous Incidents in Rail Terminals Using Machine Learning"** has been carried out by me under the supervision of Internal Guide **PROF. BARNALI CHAKRABORTY, Associate Professor, Department of MCA, AMC Engineering College**, submitted in partial fulfillment of the requirements for the award of the degree of **MASTER OF COMPUTER APPLICATIONS** by **Visvesvaraya Technological University**, **Belagavi** during the academic year 2024-2025. This report has not submitted to any other Organization/University for any award of degree Certificate.

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Firstly, I thank God for helping me throughout this project work and strengthening me to complete this work successfully and at my best. I would like to dedicate this work to my parents for their constant support and encouragement during the course of this work.

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SOUJANYA B

(1AM23MC056)

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ABSTRACT

Railroad operations for both passenger and freight transport must remain reliable, accessible, well-maintained, and safe (RAMS). However, in many urban regions, railway station accidents have emerged as a critical safety concern, leading to injuries, public anxiety, reputational damage, and financial losses. Increasing passenger demand further strains infrastructure, highlighting the urgent need for robust safety management systems.

This study proposes the use of advanced artificial intelligence techniques, particularly unsupervised topic modeling with an optimized Latent Dirichlet Allocation (LDA) model, to analyze fatal accident data from the Rail Safety and Standards Board (RSSB), covering 1,000 incidents in UK railway stations. The methodology systematically identifies accident characteristics, contributing factors, and high-risk locations, supporting data-driven safety measures and risk management strategies.

By leveraging intelligent text analysis and big data analytics, the system extracts valuable insights from historical accident reports, identifies root causes, and enhances risk assessment for large-scale, long-term safety planning. The approach not only achieves high accuracy in detecting accident patterns but also provides a scalable framework for broader applications in railway safety and the wider transportation industry.

Keywords: Railway Safety, Accident Analysis, Artificial Intelligence, Latent Dirichlet Allocation (LDA), Topic Modeling, Risk Management, Big Data Analytics, RSSB.